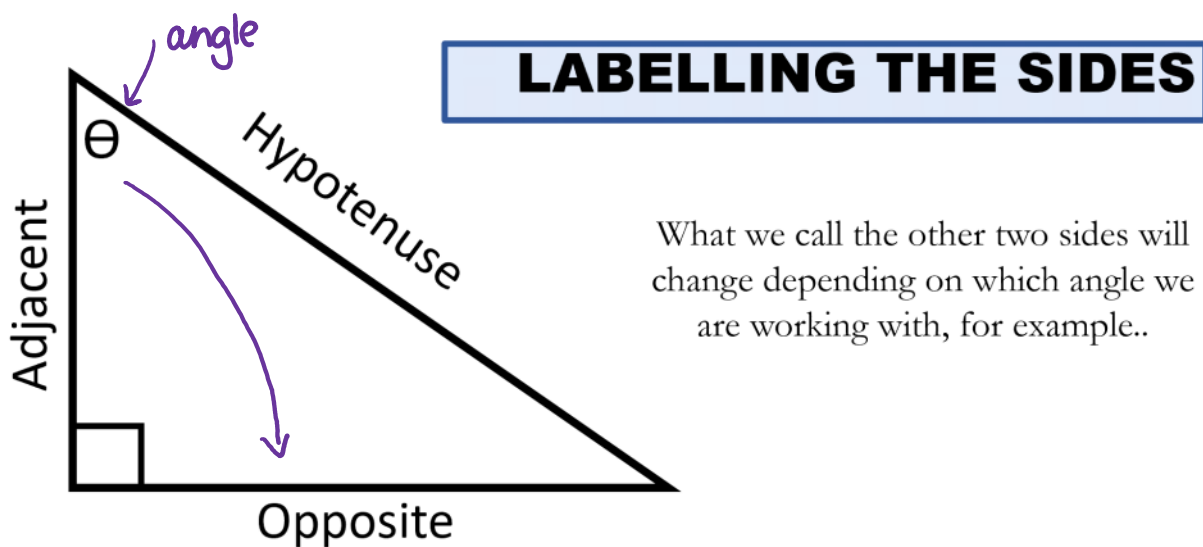
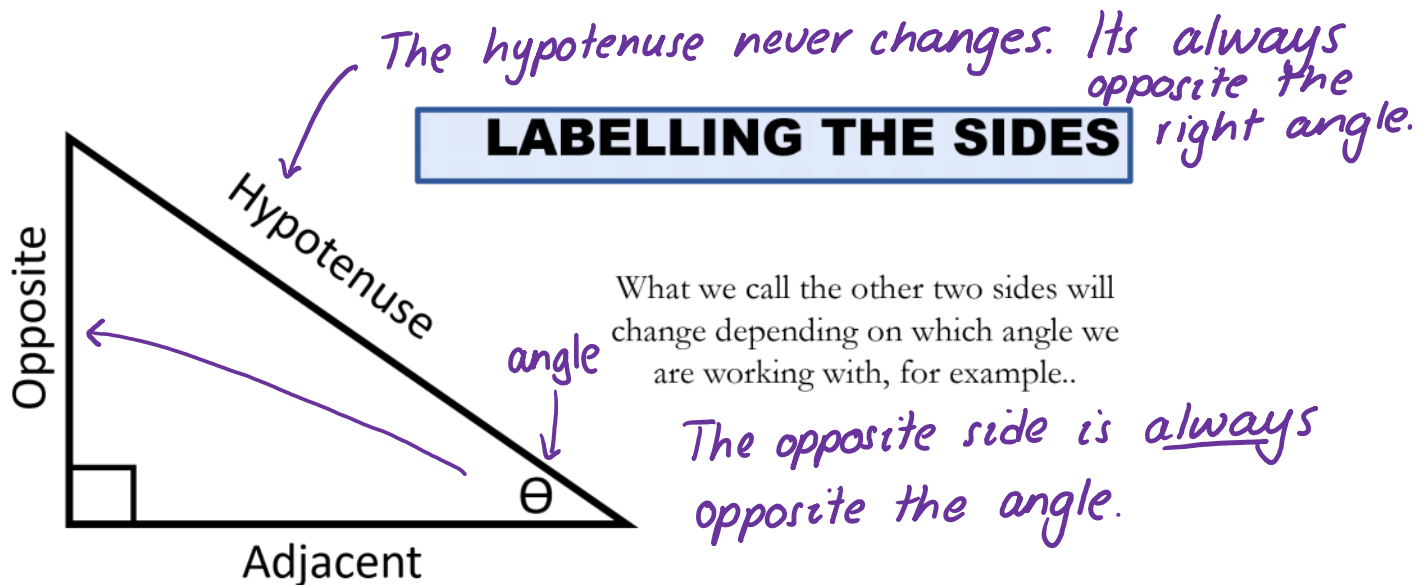
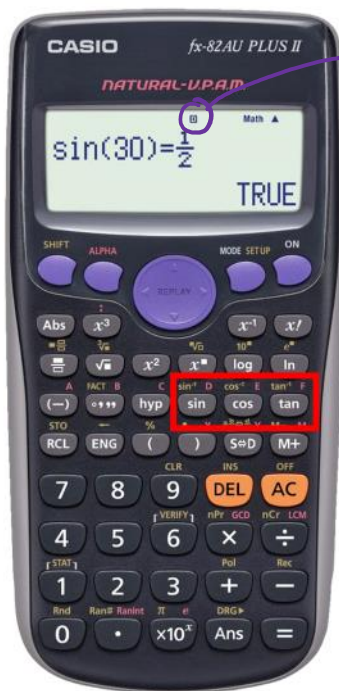


We can use trigonometry to find missing angles and lengths of triangles.

Trigonometry uses three functions, these are called:

- Sine (shortened to Sin and pronounced "sign")
- Cosine (shortened to Cos)
- Tangent (shortened to Tan)

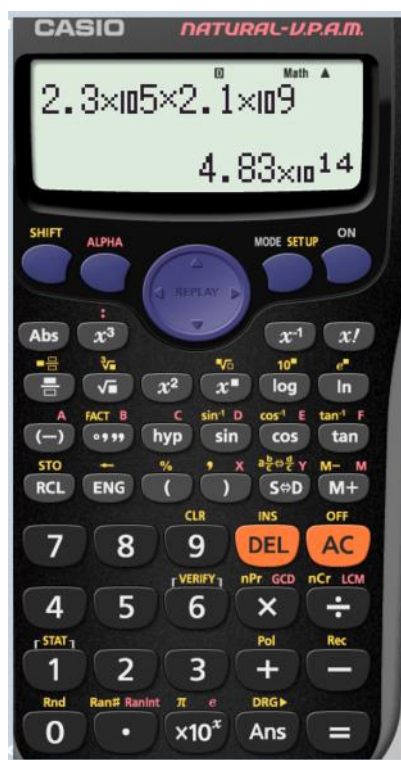




You want to see that

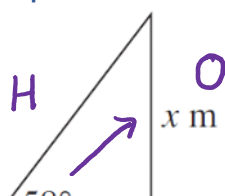
Your calculator must be set to **Deg**

- 1) Shift – mode
- 2) Select 3:Deg so press 3 on the calculator



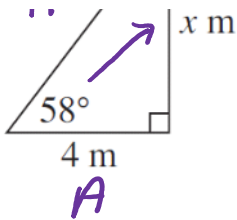
1. To get the inverse functions press Shift
2. Press tan
3. On your calculator screen it should show this : $\tan^{-1}$
4. Don't forget to close bracket once you have put in your numbers.

Example 1



$$\tan \theta = \frac{O}{A}$$

$$4 \times \tan 58^\circ = \frac{x}{4}$$

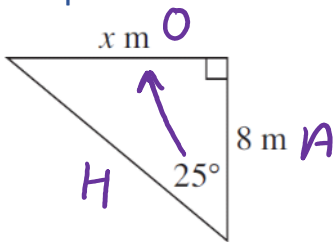


$$4 \times \tan 58^\circ = \frac{x}{4} \quad x/4$$

$$x = 4 \times \tan 58^\circ$$

$$x = 6.4 \text{ m}$$

Example 2



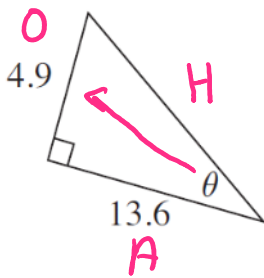
$$\tan \theta = \frac{O}{A}$$

$$8 \times \tan 25^\circ = \frac{x}{8} \quad x/8$$

$$x = 8 \times \tan 25^\circ$$

$$x = 3.7 \text{ m}$$

Example 3



$$\tan \theta = \frac{O}{A}$$

$$\tan \theta = \frac{4.9}{13.6}$$

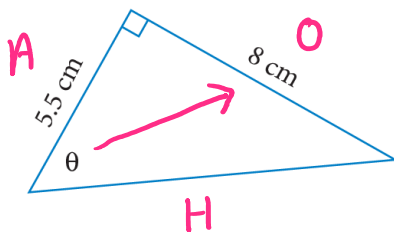
$$\tan \theta = 0.35$$

$$\tan^{-1}(4.9 \div 13.6) \quad \tan^{-1}(0.35)$$

$$\theta = 19^\circ$$

$$\theta = 19^\circ$$

Example 4



$$\tan \theta = \frac{O}{A}$$

$$\tan \theta = \frac{8}{5.5}$$

$$\tan \theta = 1.45$$

$$\tan^{-1}(8 \div 5.5) \quad \tan^{-1}(1.45)$$

$$\theta = 55^\circ$$

$$\theta = 55^\circ$$

IMPORTANT

Tangent ratio: $\tan(\theta) = \frac{\text{opposite}}{\text{adjacent}}$

To find the size of the angle when we have two sides of a right-angled triangle, we have to use the **inverse** of the trigonometric function.

The inverse of Sin is Sin^{-1}

The inverse of Cos is Cos^{-1}

The inverse of Tan is Tan^{-1}

Therefore:

For $\text{Tan } \theta = \frac{\text{O}}{\text{A}}$

Then $\theta = \text{Tan}^{-1} \left(\frac{\text{O}}{\text{A}} \right)$